

Neural means and kernel corrections for operator learning

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Abstract

A kernel fitted after a neural network can correct its residuals or replace its output layer. We study these two roles on public operator-learning problems, repeating the main experiments at ten seeds. On structural mechanics, a residual kernel correction of a six-member stack reaches $4.546 \pm 0.003\%$ mean relative test error, close to the published PARA-Net score of 4.55%, without establishing statistical equivalence. In the 1250-label regime, the pipeline reaches $5.433 \pm 0.093\%$ against a published 6.49%. On OCO-2, the same Matérn regression scores approximately 40% reduced error on raw inputs and 4.1% on frozen neural features in the O2 band. A per-coordinate combination improves on the source emulator’s stored predictions in the ten-seed means under both retained-representation rescaling criteria on all three bands; full-spectrum and retrieval accuracy are outside this comparison. Residual alignment explains the limited gains from averaging the trained mechanics predictors: a block lower bound of 4.814% lies close to the hindsight-optimal global convex RMS error of 4.876% for the sixty-member pool. Supporting ensemble identities and conditional native-space bounds clarify the scope of these measurements. Calibrated uncertainty sets are also compared; a constant-width set is narrower on average than one scaled by the kernel power function. The public test blocks were examined across several campaigns, so the results constitute retrospective benchmark evidence.

Keywords: operator learning, kernel ridge regression, residual correction, ensembles, conformal prediction

1 Introduction

A trained neural network leaves two natural places for a kernel regression: on the residual of its prediction, or on the representation learned by its hidden layers. Both constructions are well established, but their usefulness depends on the problem. We examine that dependence on two public operator-learning benchmarks, using the same basic components and repeating the main experiments at ten seeds.¹

Operator learning approximates solution maps of partial differential equations and physical forward models. Neural operators learn representations suited to these maps (Kovachki et al., 2023; Lu et al., 2021; Li et al., 2021); kernel and Gaussian-process methods offer exact finite-design solves and conditional error bounds (Battlé et al., 2024; Mora et al., 2025; Owhadi and Scovel, 2019). Neural-mean Gaussian processes (Mora et al., 2025), deep kernel learning (Wilson et al., 2016; Calandra et al., 2016), and post-hoc feature-space residual models (Vendrell-Gallart et al., 2026) combine these approaches in different

1. Code, data pointers, and per-run summaries are at <https://github.com/yspennstate/neural-means-kernel-corrections>. The current online supplement is available there as [paper/supplement.pdf](#); consult it for the full proofs. Each formal result identifies its proof location.

Table 1: The two problems studied in depth. OCO-2 rows compare against the Gaussian-process emulator of Lamminpää et al. (2025), scored from its own stored predictions on the same 2000 test points, under the two retained-representation rescoring criteria defined here (Section 5); ours are mean $\pm$ standard deviation over ten seeds per band. Structural-mechanics rows quote the best published mean relative L^2 test error at each training size (de Hoop et al., 2022; Mora et al., 2025), reported without uncertainties in the original work; ours are mean $\pm$ standard deviation over ten seeds, The 20000-sample scores are numerically close, but the available results do not establish statistical equivalence (Section 4.4).

Problem	Metric	Source predictions, rescored	This work
OCO-2, O2 band	reduced	16.89%	$4.12 \pm 0.05\%$
	radiance	0.0448%	$0.0294 \pm 0.0020\%$
OCO-2, weak CO ₂	reduced	24.06%	$16.28 \pm 0.06\%$
	radiance	0.0599%	$0.0507 \pm 0.0052\%$
OCO-2, strong CO ₂	reduced	16.14%	$8.08 \pm 0.01\%$
	radiance	0.1147%	$0.0584 \pm 0.0041\%$
Structural mechanics, 20000 samples	relative L^2	4.55% (PARA-Net)	$4.546 \pm 0.003\%$
Structural mechanics, 1250 samples	relative L^2	6.49% (FNO-mean GP)	$5.433 \pm 0.093\%$

ways. Our contribution is a controlled empirical study of where the kernel helps, how the members’ residuals limit gains from averaging, and how the evaluation metric affects the final combination. The supporting theory makes these comparisons precise without proposing a new hybrid construction.

On structural mechanics, a Matérn regression corrects the residual of a stacked neural mean. The six-member pipeline reaches $4.546 \pm 0.003\%$ mean relative test error, close to the published PARA-Net score of 4.55% (Table 1). The comparator’s predictions and retraining variability are unavailable, so the numerical proximity establishes neither superiority nor equivalence. With a budget of 1250 labels, the pipeline reaches $5.433 \pm 0.093\%$ against a published 6.49%, although the exact training subset used by the source study is not fully documented.

The structural-mechanics experiments also show why adding predictors yields diminishing returns. Residuals from different architectures are strongly aligned, and the hindsight-optimal global convex combination of sixty trained members has empirical root-mean-square relative error 4.876%. A lower bound from their within- and between-architecture second moments is 4.814% (Theorem 6.6). The narrow gap describes the remaining scope for global convex averaging in this pool. It does not constrain other architectures or the per-pixel stacks and kernel corrections used by the full pipeline. The common residual may reflect shared approximation bias, numerical properties of the deterministic target, or both; resolving its origin requires additional solver experiments.

On OCO-2, the kernel’s useful role changes. On the O2 band, the raw-input Matérn regression has approximately 40% reduced error, the flat-loss network 4.4%, and the same kernel on its frozen features 4.1%, under a common kernel grid and solve routine. The learned representation makes a substantial difference to the fitted kernel. Finite-design RKHS norms and power functions describe aspects of that difference, while the pullback identity supplies the relevant function-space interpretation; it does not predict a gain over raw

inputs. A per-coordinate combination improves on the source emulator’s stored predictions in both mean rescoring criteria on all three bands, with the weak-CO₂ radiance comparison winning at nine of ten seeds. These criteria concern the retained PCA representation, not full-spectrum instrument-noise acceptance or atmospheric retrieval.

The experiments distinguish several kinds of evidence. Seed replication measures variability within the specified procedures, and paired evaluation measures differences on a common test block. Those public blocks have been examined across successive campaigns, so the comparisons remain retrospective. The theoretical identities describe global convex ensembles exactly; the capacity and RKHS bounds require additional assumptions that need not hold for the fitted pipeline. Split-conformal prediction gives marginal coverage under its independence and exchangeability conditions, while the coverage measurements here assess performance on the repeatedly studied blocks. Among the calibrated sets examined, the constant-width set is narrower on average than the power-function-scaled set.

Section 2 describes the data and comparison protocols. Section 3 specifies the implementation, including pooled training-target centering in the foldwise kernel channel and in-sample neural predictions in the correction labels. Sections 4–5 present the experiments, and Section 6 develops the ensemble and kernel bounds, including the correction-label mismatch term. Proofs, implementation details, and additional measurements are provided in the accompanying supplement. Section 7 considers the implications and remaining questions.

2 Benchmark, protocol, and related work

2.1 Problem and data

The dataset originates in the cost-accuracy study of de Hoop et al. (2022) and is the one distributed with Batlle et al. (2024) and Mora et al. (2025). The domain $D = (0, 1)^2$ is the unit cell of a fibre-reinforced composite: a cylindrical fibre of linear elastic material at the centre, embedded in an incompressible Rivlin–Saunders (hyperelastic) matrix, so the constitutive law is heterogeneous, with a material interface across which the stress jumps. The displacement field w satisfies $\nabla \cdot \sigma = 0$ under that law, with the bottom edge clamped and a prescribed tension traction $\bar{t}$ on the top edge $\Gamma_t = [0, 1] \times \{1\}$. The quantity of interest is the von Mises stress field σ_v on D . The learning task is the map $G : \bar{t} \mapsto \sigma_v$. Loads are drawn from the Gaussian field $\mathcal{GP}(100, 400^2(-\Delta + 3^2 I)^{-1})$ with homogeneous Neumann boundary conditions for the Laplacian. In the source study the solver is a mesh of 189 quadratic quadrilateral elements (the NNFEM library) with the stress evaluated at its Gaussian quadrature points and transferred to a regular 41×41 grid by radial basis function interpolation; the distributed arrays are those grid values, with the load sampled at 41 points. The interface, the quadrature-to-grid transfer and the finite element discretization are all part of what a surrogate is asked to reproduce. The distributed file contains 40000 input/output pairs; the input array stores the load broadcast along the second grid coordinate, which we verified is an exact copy (maximal deviation 0 across all 40000 samples), so all our methods consume the load as a vector in $\mathbb{R}^{41}$.

Errors are mean relative L^2 errors over the test set,

$$\frac{1}{N_{\text{test}}} \sum_n \frac{\|\hat{G}(u_n) - G(u_n)\|_2}{\|G(u_n)\|_2},$$

computed on grid values in double precision. Quadrature weighting (trapezoidal instead of plain vector norms) changes our reported numbers by less than 0.02 percentage points, and we report the plain-norm convention of the prior work.

2.2 Protocols and published results

Two regimes appear in the literature, and we follow both. In the *high-data* protocol the first 20000 samples are available for training and the last 20000 form the test set; de Hoop et al. (2022) report, at 20000 training samples, 4.55% (PARA-Net), 4.67% (PCA-Net), 4.76% (FNO) and 5.20% (DeepONet), and Batlle et al. (2024) report 5.18% for their optimal-recovery kernel method (Matérn/rational quadratic; 27.11% for the linear kernel) on the same task. Within the 20000-sample budget we hold out the last 1000 samples (of a fixed permutation) for model selection and stacking, and train on the remaining 19000, so no method of ours sees more than the 20000 training samples used by the baselines. In the *low-data* protocol of Mora et al. (2025), 1250 samples are available and the same 20000-sample test set is used; their table gives 8.70% (DeepONet), 6.62% (FNO), 6.95% (the kernel method of Batlle et al. 2024), 6.74% (their zero-mean GP), 7.12% (DeepONet-mean GP) and 6.49% (FNO-mean GP). In this regime we carve the validation split (250 samples) out of the 1250, so training uses 1000 samples and every choice made by the pipeline is informed by the 1250 available labels only.

2.3 The rest of the suite, and a check of the mirror symmetry

Two observations belong to the setting rather than to the results, and both are reported in Supplement S4. The same fixed recipe was run once, unchanged, on the other problems of the operator-learning suite that Batlle et al. (2024) assembled from de Hoop et al. (2022) and Lu et al. (2022). That survey is uncontrolled — single runs, no seed replication, and in two cases training splits or grids that differ from the baselines’ — so nothing is drawn from it beyond orientation: where the map is smooth and data are plentiful the kernel stage wins the recipe’s own selection and lands within a factor of two of the tuned per-problem kernels. The two problems studied in depth, with replication, are the subject of the rest of the paper. Separately, the mirror symmetry that Proposition S1.1 assumes is tested on the data rather than taken on faith: outputs of near-mirror inputs are some three times closer to one another than the inputs are, the correct reflection axis is singled out against the alternative, and the empirical mean and covariance of the training loads are reflection-invariant to within 1.1% and 1.5%.

2.4 Related work

The benchmark sits at the meeting point of three lines of work. *Neural operators* learn maps between function spaces: DeepONet (Lu et al., 2021) with its branch/trunk factorization, the Fourier neural operator (Li et al., 2021) and the broader neural-operator framework (Kovachki et al., 2023), and the reduced-basis PCA-Net and PARA-Net architectures assembled for the cost–accuracy study of de Hoop et al. (2022). These methods are fast and mesh-flexible but do not by themselves come with error control, and it is their numbers on this problem that define the plateau we start from. *Kernel and Gaussian process methods* for operator learning

approach the same maps through reproducing kernels: the optimal-recovery framework of Batlle et al. (2024), with its convergence theory and a-priori bounds, is the most direct comparison, and it is competitive with neural operators on most of the benchmarks it considers. Data-adapted kernels learned by cross-validation (Darcy et al., 2023) and vector-valued kernel formulations extend the reach of this family. *Hybrids* that place a neural network and a kernel in the same estimator are the closest relatives of our pipeline: Mora et al. (2025) use a neural operator as the mean of a Gaussian process and fit the two together, and this is the work we most directly build on, differing in that we fit the mean first and the kernel after, tune by cross-validation rather than marginal likelihood, and solve the correction exactly at nineteen thousand points. Kernels on learned representations are likewise established: deep kernel learning (Wilson et al., 2016; Calandra et al., 2016) places a Gaussian process on the features of a network and trains through it, and the recent REEF-GP of Vendrell-Gallart et al. (2026) places a Gaussian process on the residuals of a trained neural operator in its learned feature space, post hoc, for uncertainty quantification. Our feature-space stage is the post-hoc form of the same construction, with the exact solve and validation tuning used everywhere else in the pipeline; it differs from REEF-GP in what it is used for and how it is judged — as a predictor, scored against the same kernel on the raw input under one tuning routine and budget (Section 5), with coverage supplied by split-conformal calibration rather than by the posterior — and what we add is not the construction but that controlled comparison. The residual construction itself is older than any of these: it is the discrepancy model of Kennedy and O’Hagan (2000), in which a Gaussian process corrects a cheap approximation, and the reduced-order-model error surrogates of Drohmann and Carlberg (2015) put a Gaussian process on the error of a physics-based surrogate in the same way; universal kriging with a trained mean is the same estimator in geostatistical dress. Within operator learning, Kumar et al. (2024) induce a Gaussian process from a neural operator, Magnani et al. (2025) linearize a trained neural operator into a function-valued Gaussian process, and Ma et al. (2024) calibrate operator predictions by split conformal prediction, the same coverage device we use in Section 4.7. Ober et al. (2021) document how deep kernel learning overfits when the feature map and the kernel are trained together and the same data select both, which is one reason our feature-space stage trains the network first and tunes the kernel by validation afterwards.

Two further threads inform the design. The kernel-flow method (Owhadi and Yoo, 2019) learns a kernel from data by minimizing a half-sample cross-validation loss, and its use as a regularizer for the inner layers of a network (Yoo and Owhadi, 2021) is exactly the term we analyze in Proposition S1.2; kernel flows have since been used at scale as emulators for physical forward models, for instance in atmospheric radiative transfer retrievals (Lamminpää et al., 2025) and in the inference of convective-storm structure from satellite observations (Prasanth et al., 2021), where presenting the input in its physical parametrization and adapting the kernel to data are what make the emulator accurate. Our pipeline follows the same instinct, with the smoothing of the target performed by a neural ensemble rather than by a hand-chosen reduction. Finally, the effective-dimension reading of the kernel solve (Lemma S5.1) draws on the random-matrix description of kernel Gram spectra (Koltchinskii and Giné, 2000) and on the classical Marčenko–Pastur (Marčenko and Pastur, 1967) and spiked-covariance (Baik et al., 2005) laws; the same effective dimension governs the generalization of kernel ridge regression (Caponnetto and De Vito, 2007).

3 Method

The pipeline has three stages: neural means, stacking, and kernel corrections. All components operate on the load vector $u \in \mathbb{R}^{41}$ (standardized by training statistics) and produce stress fields in $\mathbb{R}^{41 \times 41}$. Neural members use either mean relative L^2 loss in original units or normalized mean-squared loss, as identified in Table 4. The normalized-MSE MLP is the stronger of the two plain MLPs in this campaign.

3.1 Neural means

Residual MLP. The primary mean is a residual multilayer perceptron: three or five residual SiLU blocks of width 1024–1536 mapping $\mathbb{R}^{41} \rightarrow \mathbb{R}^{1681}$ directly. This is essentially PARA-Net (de Hoop et al., 2022) with a modern training recipe, and it also supplies the penultimate features used by the feature-space kernel stage.

Kernel-conditioned refiner. The second mean is the same residual network given an extra input channel: the kernel method’s predicted stress field for the same load, concatenated with the load. It therefore learns to correct the kernel predictor rather than to reproduce the map from scratch; its test error is close to that of the normalized-MSE network (Table 4). During training the refiner reads four-fold kernel predictions. The kernel weights are fitted excluding the held-out fold, but the recorded runs centered targets using the mean over the full fitting pool. Consequently a held-out row can affect its kernel channel through that pooled target mean. This is training-label reuse, not strict cross-fitting. At query time the refiner receives the full-data kernel prediction. The tables describe that recorded implementation; the effect of later fold-local centering has not been rerun here.

Other instances. The mean slot is not tied to a particular architecture. We also use a Fourier neural operator (Li et al., 2021) that consumes the broadcast load as a field and an MSE-trained variant of the MLP; both are drop-in means, both enter the ensemble of Section 4.5, and their pairwise error correlations are one of the measurements of this paper. A UNet on the same field representation is the sixth member of the second campaign (Section 4.1). We additionally implement an encoder–decoder transformer (Vaswani et al., 2017; Dosovitskiy et al., 2021) that tokenizes the 41 load samples and decodes the field by cross-attention at the grid points; it is a natural further architecture family, but we were unable to train it to the level of the others on the hardware available for this study and report the ensemble without it (Supplement S9).

All means are trained with AdamW and a cosine schedule, batches of 128–256, for up to a few hundred epochs (Supplement S9 lists every hyperparameter). During training each batch is reflected with probability 1/2 (load reversed, target field reflected along the first grid coordinate); at test time each model is evaluated as $\frac{1}{2}(f(u) + Tf(Su))$. Proposition S1.1 shows the test-time step cannot increase population risk under exact target equivariance and distribution invariance (proof in Supplement S8); the empirical symmetry checks do not prove those assumptions exactly. Training augmentation is a separate heuristic whose effect is measured in the ablation. Optionally, the training loss carries a kernel-flow term βe_2 computed from the batch (Section S1.2) with a Gaussian kernel on pooled penultimate features whose log-bandwidth is trained jointly; this variant appears in the ablation.

3.2 Stacking

With M trained models $f_1, \dots, f_M$ we form the convex combination $m(u) = \sum_m w_m f_m(u)$, $w \in \Delta^{M-1}$, minimizing the validation relative error; the deployed weights start uniform and are found by a random hill climb in logit space (600 Gaussian proposals of step 0.3, decaying by 0.3% per step) on half of the 1000 held-out samples (250 in the low-data protocol), the other half judging the per-pixel variant against them. The simplex minimizer of the measured second-moment matrix of Proposition 6.1 is used in the analyses of Section 4.6, not in the deployed stack. A per-pixel variant allows the weights (plus an intercept) to vary over the grid, ridge-fitted on half the validation split and accepted only when it beats the global weights on the other half; it is the variant the final pipeline uses. Validation-selected stacking (Wolpert, 1992) is an additional fitted stage. The independent-candidate bound of Proposition S1.3 is proved in Supplement S8; its independence hypothesis is not satisfied by the reused checkpoint-selection split, and its constants do not certify negligible selection cost here.

3.3 Kernel corrections

Let M_{tr} be the training-row mean used by the recorded code. It combines foldwise kernel predictions with in-sample predictions from fitted neural members; the refiner is evaluated using its training kernel channel. Its residual labels are $R = Y - M_{\text{tr}}$. At query time the mean is m_{full} . These need not agree on training inputs. Equation (5) gives the resulting deterministic mismatch term, with its proof immediately following that display.

The stacked mean is then corrected by kernel ridge regression of its residuals. Let $X = (u_1, \dots, u_n)$ be the training loads (standardized coordinatewise), $R \in \mathbb{R}^{n \times 1681}$ the matrix of training residuals $R_i = v_i - (M_{\text{tr}})_i$, and k a Matérn-5/2 kernel $k(u, u') = \kappa(\|u - u'\|/s)$. The correction is

$$\hat{r}(u) = k(u, X) \alpha, \quad \alpha = (K + n\lambda I)^{-1} R,$$

with (s, λ) chosen on the validation split from a small grid (s as a multiple of the median pairwise distance, $\lambda \in \{10^{-6}, 10^{-5}, 10^{-3}\}$; the grid is searched on an 8000-sample subsample and the chosen pair is refit on all of X). The recorded corrected surrogate is $m_{\text{full}} + \hat{r}$. A second corrective stage repeats the construction with the kernel evaluated on deep features, the concatenated penultimate activations of the ensemble members (reflection-averaged, standardized), fitted to the training-row residuals after the first correction; it contributes a smaller improvement and is included when it helps on validation. On OCO-2 (Section 5) the kernel stage takes a different form: a direct-target kernel ridge regression on the network's frozen last-layer features, tuned and refit by the same routine, in place of a residual correction of the network; the two uses are named separately where they appear. At $n = 19000$ the exact solve is a single Cholesky factorization of a 19000×19000 matrix, under half a minute in double precision on a laptop CPU (measured in Section S6.4), and prediction is two matrix products; no inducing points, preconditioners, or stochastic solvers are involved. The Gaussian process reading of the same formulas supplies the posterior standard deviation $P_\lambda(u)$ of (4) by a triangular solve per evaluation batch, which is the design factor in the residual bound of Theorem 6.9 and the uncertainty signal studied in Section 4.7.

Two remarks on scope. Nothing in the construction uses properties of this particular PDE beyond its assumed reflection symmetry, so the recipe (native input parametrization,

symmetrized accurate means, validated kernel correction of the residual, a conformal band built on the posterior standard deviation) can be applied to other operator-learning problems with low-dimensional input parametrizations. And the pipeline degrades gracefully: dropping the feature stage, the stacking, or the corrections recovers progressively simpler methods whose individual numbers appear in the ablation table.

4 Results on the structural-mechanics benchmark

4.1 What is replicated, and at how many seeds

The high-data experiments use ten seeds. Each seed redraws the validation curve from the 20000-row training block and reinitializes the neural members; the 20000-row public test block stays fixed. The first campaign uses a kernel predictor, a metric-loss MLP, a normalized-MSE MLP, a kernel-conditioned refiner, and an FNO whose 200-epoch schedule was interrupted by a task limit and whose retained checkpoints lie between epochs 10 and 70. The second campaign trains the FNO through a complete 100-epoch schedule and adds a UNet. It reproduces the other members to the printed precision. The low-data experiment also has ten seeds but keeps its 1000/250 training/validation split fixed.

Every structural result below refers to the recorded pooled-centering procedure of Section 3. The later fold-local-centering implementation has not supplied replacement results. Model-level selection uses validation, but the public test block was inspected across both campaigns and later diagnostics; the reported comparisons are retrospective.

4.2 Main comparison

Table 2 quotes published high-data scores and the two recorded pipelines. Our kernel reproduces the published kernel score approximately (5.197% versus 5.18%), supporting broad protocol agreement without proving all implementation choices identical. The six-member pipeline records $4.546 \pm 0.003\%$, against PARA-Net’s published 4.55%. This is numerical proximity, with no superiority or equivalence test because the comparator’s predictions and retraining variability are unavailable. The low-data result is $5.433 \pm 0.093\%$ versus a published 6.49% (Table 3); the exact source-study subset of 1250 labels is not documented identically.

4.3 Members and the stack, across seeds

Table 4 retains the stage-by-stage accounting. In the first campaign, the normalized-MSE network has the lowest single-member test mean (4.712%), followed by the refiner and the early-stopped FNO; the metric-loss MLP is less accurate. Adding the spectral member to the four-member pipeline reduces mean error by 0.039 ± 0.011 percentage points, with validation preferring its admission at all ten seeds. The second campaign’s completed FNO schedule improves its individual mean by 0.072 points, but the five-member corrected pipeline changes from 4.572% to 4.578%, a paired difference of 0.006 ± 0.013 points. Increased alignment with the other members is consistent with that lack of ensemble gain.

The UNet ranks fourth of six by mean single-member test error, yet adding it reduces the corrected pipeline’s error by 0.032 ± 0.004 points. In the first campaign, equal weighting, fitted global weights, per-pixel affine weights, and kernel correction successively reduce

Table 2: Structural mechanics, high-data protocol (20000 training samples, 20000 test). Mean relative L^2 test error. Rows above the rule are published results on the same data, quoted without uncertainties because none were reported. Rows below are ours: mean $\pm$ standard deviation over ten seeds. The four-member row is the same pipeline with the FNO removed, kept because the FNO’s admission is a measurement rather than a design choice (Section 4.3); the last two rows are the second campaign of Section 4.1, with the FNO trained to the end of its schedule and a UNet added. TTA denotes reflection averaging at test time.

Method	Error (%)	Parameters
DeepONet (de Hoop et al., 2022)	5.20	–
FNO (de Hoop et al., 2022)	4.76	–
PCA-Net (de Hoop et al., 2022)	4.67	–
PARA-Net (de Hoop et al., 2022)	4.55	–
Optimal-recovery kernel (Batlle et al., 2024)	5.18	–
Kernel ridge on loads (ours)	5.197 ± 0.003	–
Residual MLP + reflection TTA	4.836 ± 0.018	4.9M
+ affine stack over five members	4.595 ± 0.008	22.9M
+ residual kernel correction	4.572 ± 0.010	22.9M
the same pipeline without the FNO	4.611 ± 0.005	16.5M
+ complete FNO schedule and UNet, affine stack over six	4.567 ± 0.004	27.3M
+ residual kernel correction	4.546 ± 0.003	27.3M

mean error by 0.064 ± 0.007 , 0.021 ± 0.003 , 0.032 ± 0.005 , and 0.024 ± 0.005 points; each improvement occurs at ten of ten seeds. These are paired retrospective ablations. They show the behavior of this trained pool, without making a guarantee about architecture changes generally.

4.4 Which uncertainty is the binding one

The seed standard deviations measure training and selection variability on a fixed test block. Separately, the observed per-case standard deviation 0.0169 gives an estimated standard error of about 0.012 percentage points for the mean on 20000 i.i.d. cases, conditional on a fixed predictor. These are different quantities. Treating an unavailable comparator as if it had our per-case variance and zero covariance produces only a sensitivity calculation, not an estimated uncertainty for a score difference. It cannot establish either significance or equivalence to PARA-Net. Within-study comparisons can instead use paired per-case differences, and their seed spreads describe sensitivity to retraining. Repeated study-level inspection of the test block remains a limitation of both summaries.

4.5 Architectural diversity and the shared residual

The reported centered residual correlations range from 0.82 to 0.98 across the six-member campaign, with mean 0.918 ± 0.005 . These correlations describe alignment. The exact

Table 3: Low-data protocol (1250 training samples, 20000 test). Published rows from Mora et al. (2025). Our pipeline sees only the 1250 labels (250 of them used as the validation split). The pipeline row is mean $\pm$ standard deviation over ten seeds. Which stage the validation split selects varies with the seed — the input-space kernel correction at six seeds, its two-stage form at three, the feature-space one at the remaining seed — so the row is the error of the selected pipeline, not of a fixed final stage. The kernel-ridge row is the multiscale construction of Supplement S9, which has no random component under this protocol and therefore no seed spread.

Method	Error (%)
DeepONet	8.70
GP, optimal recovery (Batlle et al., 2024)	6.95
Zero-mean GP (Mora et al., 2025)	6.74
FNO	6.62
FNO-mean GP (Mora et al., 2025)	6.49
Kernel ridge on loads (ours, multiscale)	6.66
Full pipeline (ours, ten seeds)	5.433 ± 0.093

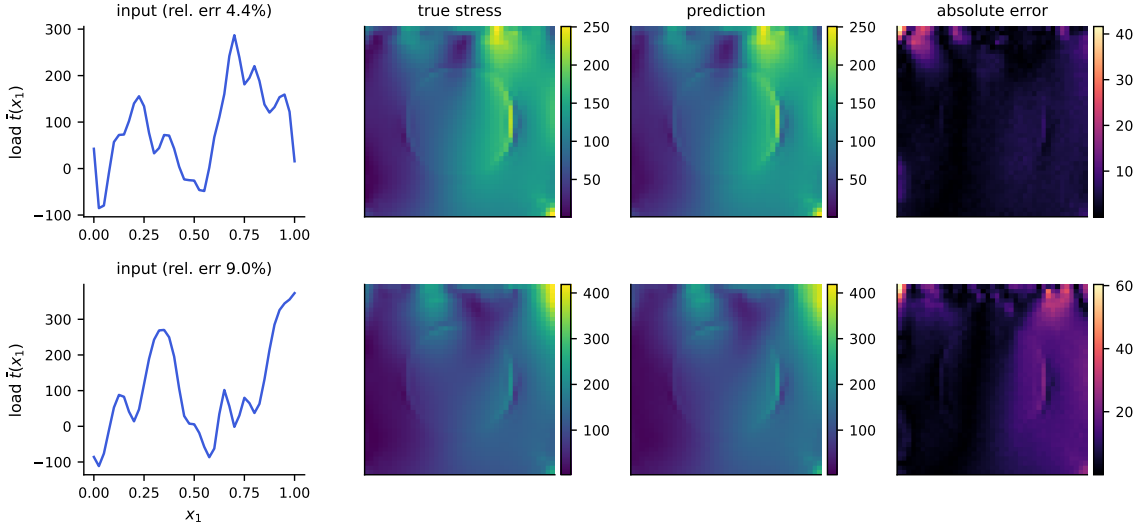


Figure 1: Example predictions of the full pipeline. Top: a median-error test case; bottom: a case at the 98th error percentile. Columns: input load, true von Mises stress, prediction, and pointwise absolute error (note the smaller scale). Errors concentrate in the high-traction corner, as also reported by Mora et al. (2025).

ensemble identity uses the uncentered normalized-residual second-moment matrix S , which is a distinct quantity.

For each seed, the analysis forms S on validation and minimizes $w^\top S w$ over the simplex. The resulting validation RMS errors are $4.940 \pm 0.057\%$ for five members and $4.920 \pm 0.057\%$ for six. The factors 0.938 ± 0.003 and 0.938 ± 0.003 compare the mean and RMS errors

Table 4: Members and stages under the high-data protocol, mean $\pm$ standard deviation over ten seeds. Member errors are reflection-averaged where the member has a reflection; the kernel ridge and the refiner train with the symmetrization already applied. “Fitted” weights are solved on the validation split and read out once on test; equal weights involve no fitting. The lower block is the second campaign of Section 4.1; its kernel ridge and MLP-family members coincide with the upper block’s to the printed precision and are not repeated. The kernel ridge member and the correction each store a 19000×1681 coefficient matrix (31.9 million numbers, 256 MB in double precision) beside the networks’ weights; the accuracy–cost–storage record is in Section S6.4.

Member or stage	Family / loss	Test error (%)
Kernel ridge on loads	Matérn-5/2	5.197 ± 0.003
Residual MLP	MLP, metric loss	4.836 ± 0.018
Residual MLP	MLP, normalized MSE	4.712 ± 0.006
Kernel-conditioned refiner	MLP on $(u, \text{KRR}(u))$	4.730 ± 0.006
Fourier neural operator	spectral, metric loss, early-stopped	4.754 ± 0.031
Equal-weight ensemble	no fitting	4.648 ± 0.005
Fitted-weight ensemble	solved on validation	4.627 ± 0.006
Affine stack (per-pixel)	selected on a held-out half	4.595 ± 0.008
+ residual kernel correction	–	4.572 ± 0.010
Fourier neural operator	spectral, metric loss, complete schedule	4.682 ± 0.009
UNet	convolutional, metric loss	4.740 ± 0.030
Equal-weight ensemble, six members	no fitting	4.626 ± 0.003
Fitted-weight ensemble, six members	solved on validation	4.602 ± 0.003
Affine stack (per-pixel), six members	selected on a held-out half	4.567 ± 0.004
+ residual kernel correction	–	4.546 ± 0.003

of the same fitted mixture on that same validation set. They are dispersion diagnostics, not prospective tests of generalization. For six members, the separately recorded test-mean/validation-RMS ratio is approximately 0.936 ± 0.011 ; this ratio combines metric dispersion and transfer between samples.

The equicorrelation references 4.922% and 4.908%, formed from average centered correlations and member RMS errors, are descriptive approximations. They are not the rigorous entrywise lower bounds of Corollary 6.3. Section 6.2 gives the latter directly from the measured matrix. No RMS lower bound is compared with a mean-error table entry by reusing another predictor’s dispersion factor.

The across-member mean residual carries about 90% of average residual energy on validation and concentrates near the corners of the loaded edge. It has more high-frequency energy than the target fields, and leading residual directions align across architectures (Supplement S6). These observations establish shared structured error in the trained pool. They do not identify its cause. Common surrogate bias and finite-element or interpolation effects are possible explanations; the target is a deterministic numerical map, so solver discrepancy relative to a continuum solution is not by itself irreducible error for learning the numerical labels. No refined-mesh comparison has been performed.

Table 5: Sixty predictors, ten seeds of six architectures, on the evaluation half of the test block (19000 cases). Mean relative L^2 error, and the root mean square where the theory speaks in that metric. Convex weights are fitted on the disjoint calibration half (1000 cases) unless marked hindsight, where they are chosen on the evaluation half itself. Rows with $\pm$ are means over the ten seeds.

Pool and weights	Error (%)	RMS (%)
FNO, one seed	4.679	4.967
FNO, ten seeds, equal weights	4.649	4.940
Six architectures, one seed, equal weights	4.623 ± 0.003	4.916 ± 0.003
Six architectures, one seed, convex, fitted	4.600 ± 0.003	—
Sixty, equal weights	4.611	4.905
Sixty, convex, fitted	4.586	4.880
Sixty, convex, hindsight	4.581	4.876
Lower bound of Theorem 6.6 on the sixty	—	4.814
Six ten-seed means, per-pixel affine, fitted	4.556	—
Sixty, per-pixel affine, leave-one-out ridge, fitted	4.547	—
Six means and the learned kernel, per-pixel affine, leave-one-out ridge	4.546	—
Six ten-seed means, per-pixel affine and kernel correction, fitted	4.533	—
Deployed six-member pipeline, one seed	4.543 ± 0.003	—

Learning curves show modest additional gains over the tested data range: the normalized-MSE model moves from $4.799 \pm 0.003\%$ at 8000 fitting rows to $4.712 \pm 0.004\%$ at 19000. Extrapolations from this range do not prove a positive limiting error. Supplement S6 retains the learning curves, residual plots, ablations, and cost measurements.

4.6 Seeds against architectures

The sixty-member analysis uses ten seeds of six model configurations, treating these configurations as blocks. The public test block is split into 1000 calibration cases for fitted weights and 19000 evaluation cases; hindsight optima are explicitly computed on evaluation. The RMS-optimal global convex mixture records 4.876% RMS error and 4.581% mean error. The latter is not a minimum of the mean-error objective, which was not solved directly.

Theorem 6.6, applied to the evaluation matrix and its smallest within- and between-block entries, gives 4.814% RMS as an empirical lower bound for every global convex combination of these sixty predictors on these cases. The gap to the recorded optimum is approximately 0.063 percentage points. The bound does not extend to untrained predictors without new entrywise assumptions, and it does not cover per-pixel affine or kernel-corrected estimators.

The deployed six-member corrected pipeline has mean error $4.543 \pm 0.003\%$ on the same evaluation cases. This is below the mean error of the RMS-optimal convex mixture, not a demonstrated bound below every mixture in mean error. The pipeline fitted to the six ten-seed mean predictors records 4.533%; its gain is an empirical ablation outside the theorem’s estimator class. Table 5 preserves the additional fitted-stack comparisons and clearly distinguishes the two metrics.

4.7 Uncertainty quantification

The single-mean power-function bound and its correction-label mismatch extension appear in Section 6.4. Because the RKHS norm is unknown and the mismatch is unmeasured, neither is an evaluated numerical certificate for this pipeline. Fitted-function norms are nevertheless computable diagnostics: the squared-norm ratio is about 4900 before normalizing target energy and about 8.8 after it (Table S6.2).

The power function has weak correlation with the absolute error (0.055) and negative correlation with relative error, while correlating with output norm (0.65). Disagreement is similarly weak as an error indicator (0.074 ± 0.006 against absolute error). Agreement among predictors cannot reveal an error component shared identically by all of them, and neither scale ranks the recorded error well here.

For the six-member pipeline, conformal calibration on 1000 cases and evaluation on the disjoint 19000 give $89.9 \pm 0.8\%$ coverage at nominal 90% and $95.0 \pm 0.6\%$ at nominal 95%. Under the frozen-predictor i.i.d. continuous-score assumptions of Remark S3.2, the corresponding central 95% reference intervals are [88.0%, 91.8%] and [93.5%, 96.3%]; all ten recorded seeds lie inside each. These are conditional-model reference intervals, not evidence that study-level adaptation satisfies the theorem’s independence assumptions.

A constant-width conformal set gives $90.4 \pm 0.8\%$ and $95.4 \pm 0.5\%$ coverage and is narrower on average: the power-function set has mean width 1.11 times as large. The power-function set also varies markedly across deciles, from 69% to 99.9% coverage at seed zero. Thus marginal coverage is compatible with poor local coverage and weak error ranking. Supplement S6 includes all three scales and the five-member comparisons. The formal guarantee is stated with its hypotheses and proof pointer in Section 6.5.

5 The OCO-2 radiative-transfer emulator

The structural-mechanics benchmark put the neural mean and the kernel in the balanced regime. To see the other regime we apply the same components to the radiative-transfer emulation problem of Lamminpää et al. (2025): per spectral band of the OCO-2 instrument, learn the map from a reduced atmospheric state (20–24 dimensions after the dimension reduction of that paper) to the reduced radiance (40 PCA coefficients of the monochromatic spectrum). The OSF project (u2t8a) publishes a training pool of 20000 pairs and, separately, 2000 test states together with the kernel-flow emulator’s own predictions on them; the released overlap check reports that the two sets share no point. The recorded runs split the pool 18000/2000 into training and validation. Within each run, the validation split selects network checkpoints, kernel length scales and nuggets, and the coordinate winners below. The public test states are used for reported evaluation and subsequent diagnostics. Because this study was developed through repeated analyses of a public benchmark, its comparisons are retrospective, not a preregistered confirmatory test. This revision retains the recorded runs and introduces no new training experiments. The carve is taken from the source study’s Sobol training design while the public test states are a separate draw, and the two differ: the flat network’s mean relative error is 6.7% on the carve against 4.4% on the test block on O2, so every selection made on the carve is a selection under a shift between the two, not a same-distribution choice. The comparison is therefore computed on identical test points against the published emulator itself rather than against our reimplementations of it.

The two error criteria are defined here on the released reduced representation. Neither is the source study’s instrument-noise or retrieval acceptance test. Let $z \in \mathbb{R}^{40}$ contain the standardized coefficients, let U have the retained orthonormal PCA columns, and put $D_z = \text{diag}(s_z)$. The stored means and scales reconstruct

$$R(z) = \mu_y + U(\mu_z + D_z z), \quad U^\top U = I. \quad (1)$$

The two reported mean relative errors are

$$E_{\text{red}} = \frac{1}{N} \sum_{i=1}^N \frac{\|\hat{z}_i - z_i\|_2}{\|z_i\|_2}, \quad (2)$$

$$E_{\text{rad}} = \frac{1}{N} \sum_{i=1}^N \frac{\|D_z(\hat{z}_i - z_i)\|_2}{\|R(z_i)\|_2}. \quad (3)$$

Orthogonality gives $\|R(\hat{z}) - R(z)\|_2 = \|D_z(\hat{z} - z)\|_2$; the reconstruction offsets cancel in the numerator but remain in the denominator. The radiance criterion compares two forty-component reconstructions, so it excludes PCA truncation error, the instrument line shape, and the measurement-noise model. Its numerator weights concentrate on the leading coefficients.

For fixed nonnegative weights ω_i, a_j , the squared objective $N^{-1} \sum_{i,j} \omega_i a_j (\hat{z}_{ij} - z_{ij})^2$ separates by coordinate. Ordinary coordinatewise RMSE selection minimizes this objective with $\omega_i = a_j = 1$. Squared relative coefficient error instead uses $\omega_i = \|z_i\|_2^{-2}$; squared relative radiance error uses $\omega_i = \|R(z_i)\|_2^{-2}$ and $a_j = s_{z,j}^2$. Fixed per-sample denominators therefore preserve separability for the squared objectives, although they can change the winning member. The square root in each summand of (2)–(3) couples coordinates, so coordinatewise selection need not minimize either reported mean norm, or both simultaneously.

Table 6 and Figure 2 show the principal comparisons. The tested kernel routine scores 40% on raw input, 30% after rescaling each input coordinate by its estimated sensitivity, and 4.1% on the trained network’s features. Under this routine, learned features are more useful than the raw coordinates or the tested input rescaling. The deep kernel head is the strongest single model tested on the O2 reduced metric. Its margin over the network it reads out is $4.11 \pm 0.05\%$ against $4.43 \pm 0.05\%$, and the head has lower error at all ten recorded seeds on each band.

A linear ridge regression refitted on the same frozen features (Table 7) is worse than the network’s trained output layer on the flat-loss network at every band-seed. The Matérn head is better than both at every band-seed, with a margin over ridge of 0.60 ± 0.07 percentage points on O2 and about a quarter of a point on the other bands. Thus this tested linear-refit control does not reproduce the Matérn improvement; it does not rule out other readout objectives or optimization procedures. Adding the ridge readouts to coordinate selection changes the combination by at most 0.002 percentage points on any band.

Three statements of scope belong next to these numbers. The combination row is produced by the unweighted selector over the three networks and their three feature-kernel heads. It takes each reduced coordinate from the model with the smallest validation root-mean-square error on that coordinate; the campaign script also runs a selector weighted by the inverse squared norm of each validation state, the exact optimizer of the squared relative

Table 6: OCO-2 O2 band, scored on the 2000 public test states, which are disjoint from the training pool; within-run selection used a validation split of the pool. Every fitted row is mean $\pm$ standard deviation over ten seeds; neural networks use a matched 250-epoch budget. The kernel-flow row is the emulator of Lamminpää et al. (2025), scored from its own stored predictions on fixed test points, and is identical at every seed. “Features” means the penultimate activations of the trained network; the kernel head is the same exact Matérn solve as everywhere else in this paper, and the two raw-input rows are fit by that same kernel-tuning routine and budget as the feature heads: the same scale grid $\{0.5, 1, 2, 4\}$ times the median distance, the same nugget grid $\{10^{-8}, 10^{-6}, 10^{-4}\}$, the same 6000-sample tuning subsample, and the winner refit on all 18000 training points. This matches the kernel stage, not total computation: feature heads additionally require neural-network training. The raw-input rows’ selected settings vary from seed to seed; every feature head, by contrast, selected the largest multiplier (4) and the smallest nugget (10^{-8}) of that grid at all thirty band-seeds, so the feature-head numbers describe this finite search grid and do not establish an optimum over wider grids. The ARD radiance summary includes a bimodal selection outcome: configurations selected on reduced validation error give 0.150% radiance error at seven seeds and 0.329% at three. Entries are aggregated from the original campaign records; the ARD radiance mean and standard deviation are included here explicitly.

model	reduced	radiance
Matérn kernel, raw input, one length scale	$40.57 \pm 0.10\%$	$0.233 \pm 0.003\%$
Matérn kernel, raw input, sensitivity-scaled (ARD)	$29.98 \pm 0.15\%$	$0.204 \pm 0.087\%$
kernel-flow emulator (Lamminpää et al., 2025)	16.89%	0.0448%
residual MLP, flat metric	$4.43 \pm 0.05\%$	$0.195 \pm 0.014\%$
residual MLP, weighted-coefficient loss	$49.32 \pm 0.40\%$	$0.0442 \pm 0.0044\%$
residual MLP, exact radiance loss	$59.02 \pm 0.76\%$	$0.0571 \pm 0.0057\%$
Matérn head on the flat network’s features	$4.11 \pm 0.05\%$	$0.0793 \pm 0.0072\%$
Matérn head on the weighted network’s features	$21.70 \pm 0.52\%$	$0.0319 \pm 0.0020\%$
Matérn head on the radiance network’s features	$20.48 \pm 0.61\%$	$0.0379 \pm 0.0037\%$
per-coordinate combination	$4.12 \pm 0.05\%$	$0.0294 \pm 0.0020\%$

error that Proposition S2.2 covers, and it chooses different winners at three seeds on O2 and four on WCO2 yet changes each band’s mean reduced error by less than 0.001 percentage points, with unchanged radiance means at the reported precision (both records are in the release). On O2 the unweighted combination has mean reduced error 4.11510%, slightly above the flat-feature head’s 4.11080%, while improving radiance error from 0.07927% to 0.02940%. It therefore does not dominate every individual model on both criteria. The uncertainty of the comparison against the emulator is paired: on the same 2000 test states, a case-resampling bootstrap of the difference between the combination and the emulator excludes zero at ten of ten seeds on both metrics on O2 and SCO2, and at ten of ten on WCO2’s reduced metric and nine of ten on its radiance metric, the exception being the seed that loses by 0.0015 points; the seed standard deviations in Table 6 describe retraining variability, not that comparison. These are descriptive case-resampling intervals for the recorded predictors, with no adjustment for repeated benchmark analyses. The claim is

Table 7: Readout control for the kernel heads, ten seeds per band, mean $\pm$ standard deviation over seeds. For the flat and weighted-coefficient networks the same frozen last-layer features are read out three ways: by the network’s own trained output layer, by a ridge regression refit on those features with the penalty chosen on validation, and by the exact Matérn head. The rows come from the recorded follow-up run of the campaign script with the ridge rows added (`campaign/jpl_seeded.py`; body generated by `campaign/gen_oco_ridge_table.py`); the rerun’s network and head rows reproduce Table 6 within the seed spread.

model	reduced	radiance
<i>O2 band</i> (10 seeds)		
network, flat loss	$4.43 \pm 0.05\%$	$0.195 \pm 0.014\%$
ridge readout on its frozen features	$4.71 \pm 0.07\%$	$0.228 \pm 0.014\%$
Matérn head on its frozen features	$4.11 \pm 0.05\%$	$0.0793 \pm 0.0072\%$
network, weighted-coefficient loss	$49.33 \pm 0.40\%$	$0.0439 \pm 0.0031\%$
ridge readout on its frozen features	$45.71 \pm 0.61\%$	$0.0521 \pm 0.0043\%$
Matérn head on its frozen features	$21.72 \pm 0.52\%$	$0.0316 \pm 0.0019\%$
<i>WCO2 band</i> (10 seeds)		
network, flat loss	$16.44 \pm 0.06\%$	$0.226 \pm 0.012\%$
ridge readout on its frozen features	$16.51 \pm 0.06\%$	$0.252 \pm 0.017\%$
Matérn head on its frozen features	$16.28 \pm 0.06\%$	$0.116 \pm 0.007\%$
network, weighted-coefficient loss	$62.57 \pm 0.56\%$	$0.0738 \pm 0.0057\%$
ridge readout on its frozen features	$61.32 \pm 0.43\%$	$0.0842 \pm 0.0067\%$
Matérn head on its frozen features	$48.67 \pm 0.45\%$	$0.0533 \pm 0.0054\%$
<i>SCO2 band</i> (10 seeds)		
network, flat loss	$8.23 \pm 0.02\%$	$0.199 \pm 0.015\%$
ridge readout on its frozen features	$8.33 \pm 0.03\%$	$0.236 \pm 0.032\%$
Matérn head on its frozen features	$8.08 \pm 0.01\%$	$0.109 \pm 0.007\%$
network, weighted-coefficient loss	$36.46 \pm 0.65\%$	$0.0751 \pm 0.0025\%$
ridge readout on its frozen features	$40.10 \pm 0.91\%$	$0.0866 \pm 0.0039\%$
Matérn head on its frozen features	$15.47 \pm 0.47\%$	$0.0631 \pm 0.0037\%$

confined to what is rescored: two criteria defined here on the released reduced representation (the coefficients, and the radiance reconstructed from the retained forty components), not the full-spectrum acceptance test of Lamminpää et al. (2025), at a 250-epoch budget the networks have not converged at, with the emulator fitted on 20000 states against our 18000 plus a 2000-state validation carve. Instrument response, Jacobians and the retrieval itself are not evaluated, so nothing here says the pipeline is a better OCO-2 emulator end to end. Learning the metric rather than setting it by hand does not change this reading. A per-dimension metric fit by the kernel-flows cross-validation loss of Owhadi and Yoo (2019) reaches the same 30% on the raw state, and a low-rank Mahalanobis metric with a thousand more parameters does not improve on it; on the features, where the network has already conditioned the representation, a learned metric gives no gain over the isotropic kernel. These observations concern the tested metrics, feature maps, and tuning budgets; they do not exclude stronger raw-input kernels.

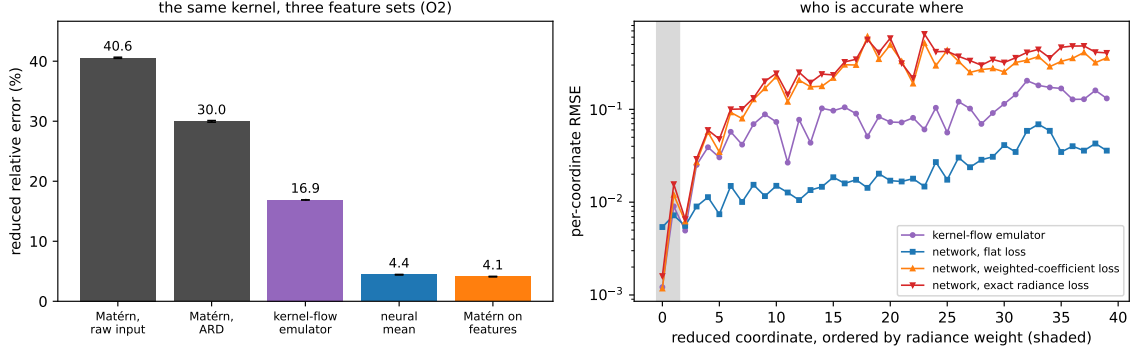


Figure 2: Both panels are drawn from the ten-seed campaign by `campaign/make_oco2_fig.py`: bars and error bars in the left panel are the mean and standard deviation over the ten seeds, curves in the right panel are per-coordinate test RMSE averaged over the same ten. Left: the representation ladder on the O2 band. The same exact Matérn machinery moves from 40% on the raw input to 4.1% on the trained network’s features; the kernel-flow emulator and the network sit in between. Right: per-coordinate test RMSE with the reduced coordinates ordered by their radiance weight (shaded). Networks are named by the loss they were trained on. The per-output-tuned kernel-flow emulator is most accurate exactly on the heavily weighted leading coefficient (0.0012 against the flat network’s 0.0054), while the flat network is nearly four times more accurate across the trailing thirty coordinates that the radiance barely sees (0.0276 against 0.1037). Both weighted losses buy the leading coefficient at the price of the tail, and by almost the same amount (0.0012 and 0.0016), which is the measurement behind the reading below; the per-coordinate combination takes each coordinate from whichever model wins it.

5.1 What the metric buys, measured

The recorded sweep refits both raw-input kernel rows at five nested training sizes, ten seeds per band, estimating the sensitivity map from the available training rows at each size. At $n = 18000$ the isotropic-to-adapted error ratios range from 1.1 to 1.5. The fitted log-log slopes differ little on O2 and more on strong CO₂; the numerical results are in Section S7. These slopes summarize a finite range of training sizes and are not asymptotic rates. Proposition S2.5 instead gives a conditional finite-design bound for rank truncation; this sweep does not verify its Lipschitz or native-space assumptions.

5.2 Retrospective transfer of the two-member rule

Part (i) of Proposition 6.1 says that a weaker member helps the optimal two-member squared-relative-error mixture exactly when $\varrho < e_1/e_2$. We examine this criterion retrospectively using stored validation and test predictions for eight models: the two raw-input kernels, three networks, and three feature-kernel heads. There are $\binom{8}{2} = 28$ pairs at each of thirty band-seeds, 840 in all. This diagnostic is not a preregistered member-admission experiment.

Evaluating both sides on the same block, the criterion agrees with whether the optimal convex mix beats the better member on all 840 pairs. This checks an algebraic identity against stored predictions; it is not independent statistical evidence for the proposition.

The separate transfer question uses ϱ and e_1/e_2 estimated on validation, then compares that verdict with the test-block result. The test-side outcome is whether the optimal convex two-member mix beats the better of the two members *on test*, whichever that is (the order of a pair flips between the splits in 40 of the 840). The rule is correct on 62.4% of the 840 pairs. Stratified by the observed margin $|\varrho - e_1/e_2|$ that the rule must resolve, the outcome is 101 of 257 below 0.02, 95 of 215 from 0.02 to 0.05, 81 of 88 from 0.05 to 0.10, 215 of 248 from 0.10 to 0.25, and 32 of 32 above 0.25 — that is 39%, 44%, 92%, 87% and 100%. That scores the existence of a helpful mix. Scoring what the rule would ship — the mix at the weight fitted on validation, against the better test member — the counts are 142 of 257, 107 of 215, 76 of 88, 125 of 248 and 31 of 32 (57.3% overall). Those are rates at which the rule’s yes-or-no prediction about the shipped outcome is right; in absolute terms 356 of the 840 transferred mixtures beat the better test member, all of them among the 715 pairs with a positive admission verdict. A verdict about the existence of a helpful mixture is different from realizing that gain using a weight transferred from validation. Every quantity here is the root-mean-square relative error and the uncentered alignment the proposition is stated in (`oco_ensemble_recheck.json` in the release). These counts are not independent observations: the 28 pairs of a band-seed share members, and the ten seeds of a band share one test block, so the bins should be read as describing a trend across margin and not as binomial samples. Nothing about the proposition degrades in the small-margin pairs; what degrades is the transfer of two estimates from one block to another.

The signed gap $D = \varrho - e_1/e_2$ changes between validation and test with empirical standard deviation 0.099 across these dependent pairs and mean change 0.0004. Its within-pair seed scatter is 0.010. The validation-to-test design shift is therefore material in this diagnostic. Proposition S2.1 would require a proved estimation-error condition to supply a probabilistic bound. The empirical standard deviation does not establish such a condition. Neither the observed margin bins nor the 32 correct existence decisions above margin 0.25 establish a validated admission threshold for future models or data.

For the raw-input kernel against the flat network, the ten-seed validation means are $\varrho = 0.156$ and $e_n/e_k = 0.157$. The difference changes sign across seeds, giving an admission verdict at five of ten. The corresponding optimal validation RMS relative errors are 8.4079% for the mixture and 8.4081% for the network, at a network weight of 0.9999. The test-block hindsight optimum differs from the network by 0.0004 percentage points. These recorded gains are too small to support a practical improvement from this pair. They explain its limited contribution retrospectively, without claiming a reliable validation-only exclusion threshold or an improvement in the reported mean-relative-error objective.

5.3 Data scaling

The recorded OCO-2 error decreases over the training sizes examined. The finite-range learning curves, and the retained large- n ClimSim records after withdrawal of leaked small- n points, are reported in Section S7. They do not identify an asymptotic rate or exclude other error sources.

6 Theory

Throughout this section $\mathcal{U} = \mathbb{R}^p$ denotes the (discretized) input space and $\mathcal{V} = \mathbb{R}^q$ the output space, $G : \mathcal{U} \rightarrow \mathcal{V}$ the target operator, μ the input distribution, and $(u_1, v_1), \dots, (u_N, v_N)$ i.i.d. draws with $v_n = G(u_n)$; the data carry no sampling noise in the usual sense, coming from a deterministic solver, though Section 4.5 discusses solver error. Assume $\|G(u)\|_2 > 0$ almost surely and that all displayed expectations are finite. We write $\ell(w, v) = \|w - v\|_2 / \|v\|_2$ for the relative error and $R(\hat{G}) = \mathbb{E}_{u \sim \mu} \ell(\hat{G}(u), G(u))$ for the risk, which is the quantity reported in Section 4. The results below are largely elementary, but each one motivates a specific design decision in the pipeline of Section 3, and each has a measurable footprint in the experiments: one statement per stage, from the symmetry handling and the kernel-flow term through stacking, the residual correction, its coverage guarantee, and the choice of nugget. Proofs are collected in Supplement S8.

6.1 Guarantees for the fitted stages

Every fitted stage of the pipeline has an elementary finite-sample bound, stated and proved in the online supplement (Section S1). Their status should be read exactly: they are conditional capacity calculations for candidates fixed before the validation sample is drawn, and the pipeline as deployed does not satisfy that hypothesis, because the same validation split that scores the stacks and the kernel grids also selects the network checkpoints, the member set and, on OCO-2, the coordinate winners. We therefore present them as the scaling behind each design decision, with constants measured on the runs, and not as guarantees for the shipped estimator; as Section S1 notes, several are numerically vacuous at the campaign’s sizes in any case. Split-conformal coverage has a different independence requirement: the predictor and scale must be fixed independently of the calibration and evaluation data. The retrospective experiment is interpreted under that requirement, not as protection against unrestricted reuse of its public test block. The bounds are: a symmetrization inequality behind the reflection averaging (Proposition S1.1), the leave-half-out reading of the kernel-flow regularizer (Proposition S1.2), capacity bounds for the validated stack, the honesty comparison, the per-coordinate affine stack, and the grid-tuned correction (Propositions S1.3 and S1.5, Lemma S1.4, Corollary S1.6), the last with an analytic uniform bound on the correction weights (Lemma S1.7, $\kappa_{\text{an}} = 500$, sharp over kernels and designs). The statistical capacity bounds are reported for their scaling. Theorems 6.9 and 6.10 give deterministic conditional bounds, whose unknown norm factors prevent numerical error certification.

6.2 What the stack can achieve

Proposition S1.3 bounds weight selection under independent-validation assumptions; it does not imply that mixing improves on a member. That is decided by a single measurable matrix. For a map f write the *normalized residual* at u as $\rho_f(u) = (f(u) - G(u)) / \|G(u)\|_2$, so that $\ell(f(u), G(u)) = \|\rho_f(u)\|_2$.

Proposition 6.1 (second-moment identity for convex stacks). *Let $f_1, \dots, f_M$ be fixed maps and define $S \in \mathbb{R}^{M \times M}$ by $S_{mk} = \mathbb{E}_{u \sim \mu} \langle \rho_{f_m}(u), \rho_{f_k}(u) \rangle$. For w in the simplex Δ^{M-1} let $f_w = \sum_m w_m f_m$. Then*

$$\mathbb{E}_{u \sim \mu} \ell(f_w(u), G(u))^2 = w^\top S w,$$

so the squared-metric risk of every convex stack is determined by S , and the best achievable is $\min_{w \in \Delta^{M-1}} w^\top S w$. In particular:

- (i) (two members) if S has diagonal (e_1^2, e_2^2) with $0 < e_1 \leq e_2$ and correlation $\varrho = S_{12}/(e_1 e_2)$, mixing in the weaker member strictly helps if and only if $\varrho < e_1/e_2$, in which case the optimum is interior with value $e_1^2 e_2^2 (1 - \varrho^2)/(e_1^2 + e_2^2 - 2\varrho e_1 e_2)$;
- (ii) (equicorrelated family) if $S_{mm} = e^2$ and $S_{mk} = \varrho e^2$ for all $m \neq k$ with $\varrho \geq 0$, then $\min_w w^\top S w = e^2(\varrho + (1 - \varrho)/M)$, attained at uniform weights and decreasing to $e^2 \varrho$ as $M \rightarrow \infty$.

Proof. See Supplement S8.10.

If a member has zero squared risk, it already attains the minimum zero; ratios involving its error are not used.

Part (i) is the classical two-member minimum-variance rule, and part (ii) is the equicorrelated case of the ambiguity decomposition of ensemble learning (Krogh and Vedelsby, 1995); see Ueda and Nakano (1996) for the generalization-error form and Brown et al. (2005) for a survey. Both enter here as measurable decision tools on the estimated S .

Everything in this subsection is stated for the squared metric, and the tables of Sections 4 and 5 report a different one. The distinction is small numerically and decisive logically, so we make it explicit once and refer back to it wherever the two families of numbers meet.

Remark 6.2 (the two error metrics). For a map f write $\mathcal{E}_1(f) = \mathbb{E}\|\rho_f(u)\|_2$ and $\mathcal{E}_2(f) = (\mathbb{E}\|\rho_f(u)\|_2^2)^{1/2}$, the mean and the root mean square of the per-sample relative error. If $\mathcal{E}_1 = 0$, both errors vanish and no ratio is needed; below assume $\mathcal{E}_1 > 0$. The tables report $\mathcal{E}_1$; Proposition 6.1 is a statement about $\mathcal{E}_2$, since $\mathcal{E}_2(f_w)^2 = w^\top S w$ by definition. Writing $X = \|\rho_f(u)\|_2$ and $c_f = (\text{Var } X)^{1/2}/\mathbb{E}X$,

$$\mathcal{E}_2(f)^2 - \mathcal{E}_1(f)^2 = \text{Var } X, \quad \mathcal{E}_1(f)/\mathcal{E}_2(f) = (1 + c_f^2)^{-1/2} \leq 1,$$

with equality only if the per-sample error is almost surely constant. Two consequences are used in this paper. The floor belongs in $\mathcal{E}_2$, because only $\mathcal{E}_2$ is determined by S : the map $w \mapsto w^\top S w$ is a quadratic form in second moments, whereas $w \mapsto \mathbb{E}\|\sum_m w_m \rho_{f_m}(u)\|_2$ depends on the full joint law of the residuals. And upper bounds transfer downward for free, since $\mathcal{E}_1 \leq \mathcal{E}_2$, while lower bounds do not — so a floor read off S says nothing about a table entry without the dispersion of the same predictor. Comparing the two directly is biased in favour of the table entry by $(1 + c_f^2)^{-1/2}$, which is about six percent on the structural-mechanics stack and ten to eleven percent on the O2 objects of Section 5; the dispersion is a property of the predictor and not a constant of the problem, so a single conversion factor should not be reused across objects.

Part (i) is an exact criterion, but it is used as a decision rule and a decision rule is applied to estimates, so what governs it in practice is not the criterion but the gap it has to resolve relative to the noise in that estimation. A margin law for threshold rules of this shape (Proposition S2.1, Supplement S2) bounds the probability that a decision is both wrong and taken at an observed margin of at least τ by $\exp(-\tau^2/2\sigma^2)$, with σ the scale of the estimation error and not a chosen constant. That is a joint bound, not the error rate

among the decisions whose observed margin exceeds τ : the conditional quantity is the joint one divided by $\mathbb{P}(|\widehat{D}| \geq \tau)$, on which the proposition says nothing. Accuracy inside a margin bin is therefore an empirical calibration of the rule, and Section 5.2 reports it that way.

Exact equicorrelation never holds in practice, but for convex weights the floor survives one-sided bounds on the entries of S , which are what one actually measures.

Corollary 6.3 (ensembling floor). *If every member satisfies $S_{mm} \geq \bar{e}^2$ and every pair satisfies $S_{mk} \geq \bar{e} \bar{e}^2$ with $\bar{e} \in [0, 1]$, then every convex combination f_w obeys*

$$\mathbb{E} \ell(f_w(u), G(u))^2 = w^\top S w \geq \bar{e}^2 (\bar{e} + (1 - \bar{e}) \|w\|_2^2) \geq \bar{e}^2 \bar{e}.$$

No number of additional members with the same error level and mutual correlations moves the ensemble's root-mean-square relative error below $\bar{e} \sqrt{\bar{e}}$.

Proof. See Supplement S8.11.

The floor is a lower bound; a matching upper bound follows from the same entrywise information, and together they pin the achievable window from both sides by measured quantities alone.

Corollary 6.4 (the floor is nearly achieved). *Suppose $\bar{e} > 0$ and, in addition to the hypotheses of Corollary 6.3, that $S_{mm} \leq \bar{e}^2(1 + \delta_e)$ for every m and $S_{mk} \leq (\bar{e} + \delta_e) \bar{e}^2$ for every $m \neq k$. Then*

$$\bar{e} \leq \frac{1}{\bar{e}^2} \min_{w \in \Delta^{M-1}} w^\top S w \leq \bar{e} + \delta_e + \frac{1 + \delta_e - \bar{e} - \delta_e}{M}.$$

The width of the bracket is the measured correlation spread δ_e plus a $1/M$ term: when these entrywise spreads are small, the optimal convex risk is localized. The upper bound is obtained by evaluating uniform weights; it does not upper-bound an arbitrary fitted convex combination.

Proof. See Supplement S8.6.

Applied to a measured matrix $\widehat{S}$, these are exact empirical bounds on that same design, not confidence bounds on the population matrix. For the six-member validation matrices, the weak RMS floor $\sqrt{\min_{i \neq j} \widehat{S}_{ij}}$ is $4.820 \pm 0.062\%$, the stronger finite-six-member lower bound is $4.849 \pm 0.060\%$, and the simplex optimum is $4.920 \pm 0.057\%$ (mean and sample standard deviation over ten seeds). The values near 4.91% obtained from mean centered correlations are equicorrelation diagnostics, not these lower bounds.

The floor is stated for convex weights, and the deployed stack is not convex: its per-pixel weights are affine and unconstrained in sign. For global weights the sign constraint can be removed exactly.

Corollary 6.5 (signed stacks). *If S is positive definite, the minimum of $w^\top S w$ over all w with $\sum_m w_m = 1$, of either sign, is $1/(\mathbf{1}^\top S^{-1} \mathbf{1})$, attained at $w^* = S^{-1} \mathbf{1}/(\mathbf{1}^\top S^{-1} \mathbf{1})$; it is at most the convex minimum, with equality exactly when w^* has no negative coordinate.*

Proof. See Supplement S8.9.

On the measured S the two minima agree to within 0.0005 points at every seed and for both member sets, but not because the constraint is slack: the unconstrained minimizer

carries a small negative coordinate at seven seeds of ten for the six members and at one seed for the five (the stored **signed** records), and removing it costs at most those five ten-thousandths. Allowing signed global weights therefore buys nothing measurable here, and what the deployed per-pixel stack gains over the convex one (0.032 and 0.035 points in the two campaigns) is a benefit of a different per-pixel affine class; the global signed-weight comparison does not isolate the effect of signs or intercepts within that class.

The members of a pool are usually of two kinds at once, architectures and seeds of each architecture, and the floor separates the two.

Theorem 6.6 (seeds and architectures). *Let the pool consist of K seeds of each of M architectures, and suppose $S_{ii} \geq \bar{e}^2$ for every member, $S_{ij} \geq \varrho_w \bar{e}^2$ for two seeds of the same architecture and $S_{ij} \geq \varrho_b \bar{e}^2$ for members of different architectures, with $0 \leq \varrho_b \leq \varrho_w \leq 1$. Then every convex combination obeys*

$$\mathbb{E} \ell(f_w(u), G(u))^2 \geq \bar{e}^2 \left(\varrho_b + \frac{\varrho_w - \varrho_b}{M} + \frac{1 - \varrho_w}{MK} \right),$$

with equality at uniform weights when the three inequalities are equalities. For an extended pool only if the same entrywise bounds continue to hold, no number of seeds of one architecture takes the root-mean-square error below $\bar{e}\sqrt{\varrho_w}$, no number of seeds of M architectures takes it below $\bar{e}\sqrt{\varrho_b + (\varrho_w - \varrho_b)/M}$, and no pool of any size takes it below $\bar{e}\sqrt{\varrho_b}$.

Proof. See Supplement S8.12.

At fixed M , increasing K reduces only the term $(1 - \varrho_w)/(MK)$ in the lower bound. The within-architecture contribution $(\varrho_w - \varrho_b)/M$ persists. Section 4.6 applies the formula empirically to sixty trained predictors; it does not assume the same inequalities for untrained members.

The two-member rule of Proposition 6.1(i) can be read the same way, as a statement about what a member's accuracy is worth against its correlation, and the exchange rate is explicit.

Proposition 6.7 (the price of accuracy in correlation). *In the setting of Proposition 6.1(i) write $t = e_2/e_1$ and assume $-1 < \varrho < 1/t$ (with $t \geq 1$), so the optimum is interior. The optimal two-member risk is $V = e_1^2 t^2 (1 - \varrho^2)/(1 + t^2 - 2\varrho t)$, and*

$$\frac{\partial V}{\partial t} = \frac{2e_1^2 t (1 - \varrho^2)(1 - \varrho t)}{(1 + t^2 - 2\varrho t)^2} > 0, \quad \frac{\partial V}{\partial \varrho} = \frac{2e_1^2 t^2 (t - \varrho)(1 - \varrho t)}{(1 + t^2 - 2\varrho t)^2} > 0.$$

A change $(dt, d\varrho)$ in the second member therefore leaves the optimal risk unchanged to first order exactly when $d\varrho = -(1 - \varrho^2) dt / (t(t - \varrho))$: an improvement of the second member's error by a fraction ϵ is cancelled by a rise of $\epsilon(1 - \varrho^2)/(t - \varrho)$ in its correlation with the first.

Proof. See Supplement S8.13.

At perfect anticorrelation $\varrho = -1$, cancellation gives zero optimal risk and the derivative in t is zero, so that boundary is excluded above. At the admission threshold $\varrho = 1/t$ with $t > 1$, the exchange coefficient $(1 - \varrho^2)/(t - \varrho)$ tends to $1/t$; it need not be large. The formula is a local sensitivity calculation at an interior point. Supplement S6 evaluates it at the measured values.

For coordinate-dependent combinations, squared losses with a fixed diagonal metric separate by coordinate (Proposition S2.2; proof in Supplement S8). The same holds after a fixed sample weighting (Corollary S2.3; proof in Supplement S8). These statements motivate the selector but do not make it simultaneously optimal for the two reported mean relative norms: the square root couples coordinates, and the two criteria also have different sample denominators. Proposition S2.4 gives an independent-validation selection bound (proof in Supplement S8); the deployed candidates reuse validation, as noted above. The OCO-2 combination improves on the rescored source emulator in both mean criteria, while the flat feature head is slightly better on the O2 reduced criterion.

6.3 Metric and representation: when a fixed kernel loses to a network

The structural-mechanics benchmark has the neural mean and the isotropic kernel within half a point of each other. On the OCO-2 emulation problem of Section 5 the same fixed kernel is an order of magnitude behind the same network. Two mechanisms can produce such a gap, and they separate cleanly.

The first is input geometry. A sensitivity-based linear transformation improves the raw-input kernel by a factor of about 1.1–1.5 in this study. The learning-curve sweep does not establish a general rate improvement. Proposition S2.5 in Supplement S2 gives a finite-design bound for a Lipschitz target after a rank- r truncation, with its approximation error explicit; its proof is in Supplement S8. It makes no assertion that a measured approximate rank changes an asymptotic rate.

The second is the representation used by the kernel. Native-space bounds depend on the target’s norm in that space. A fixed kernel may approximate targets outside its RKHS, so native-space membership is not a limit on approximation expressivity. Learned features can change the relevant norm and design geometry, but neither change is automatically favorable.

The representational term has a precise reading through the same optimal-recovery bound that Section 6.4 makes exact. That bound controls the error by $\|G - m\|_{\mathcal{H}} \cdot P_{\lambda}$, a product of the target’s norm in the kernel’s native space and a design factor determined by the kernel, training inputs, and query. Changing the features changes both in principle, and on the O2 band the two move by very different amounts. The effective dimension of the Matérn Gram, the quantity Lemma S5.1 controls, is nearly identical on the raw input and on the learned features (at $n = 4000$ and a matched median length scale, 3995 against 3993 at a 10^{-8} nugget); the design factor itself is not fixed by that trace, and measured directly at the 2000 test inputs the feature kernel’s power function is smaller at the median scale, its median 1.5 times below the raw kernel’s, and equal to it (0.99) at the multiplier 4 that the campaign selected for every feature head. The fitted-norm diagnostic moves far more: regressing the same outputs through the two kernels, the squared native-space norm of the fitted interpolant, $\text{tr}(\alpha^{\top} K \alpha)$, is 38 times smaller on the features than on the raw input at the median scale (77 times on the full 18000-row block) and 273 times smaller at the selected multiplier (139 times on the full block). (An earlier draft quoted 42 from $\alpha^{\top} Y$, which adds the term $n\lambda\|\alpha\|^2$; at this nugget the two agree to three digits, and the network retrained for the check lands at 38.) Both numbers are finite-design diagnostics of the interpolating regime: for a target in the corresponding RKHS with compatible labels, the fitted norm is a

lower bound on its norm, the ratio grows with the kernel scale — 6 at half the median length scale, 154 at twice it — and at a 10^{-4} nugget, where the ridge no longer interpolates, it collapses and at the selected multiplier inverts (0.2). Both designs are standardized by their training moments before the kernel is formed, as the campaign’s heads are, and the check is at one seed. Read as diagnostics, they say that the learned features hand the same kernel a target it interpolates with a much smaller fitted RKHS norm, and a somewhat smaller pointwise design factor at the median scale; they do not identify a causal decomposition into a fixed design factor and a moving norm.

The associated feature-kernel function space has a standard pullback description. Writing φ for the feature map and k for the base kernel, the deep kernel head uses $k_\varphi(x, x') = k(\varphi(x), \varphi(x'))$, the construction studied as deep kernel learning (Wilson et al., 2016; Calandra et al., 2016).

Proposition 6.8 (feature pullback). *The native space of k_φ is $\mathcal{H}_{k_\varphi} = \{f \circ \varphi : f \in \mathcal{H}_k\}$ with $\|g\|_{\mathcal{H}_{k_\varphi}} = \min\{\|f\|_{\mathcal{H}_k} : f \circ \varphi = g \text{ on } \mathcal{U}\}$. In particular, if the target factors through the features as $G = h \circ \varphi$ with $h \in \mathcal{H}_k$, then $\|G\|_{\mathcal{H}_{k_\varphi}} \leq \|h\|_{\mathcal{H}_k}$, and the optimal-recovery bound of Theorem 6.9 for the feature-kernel regression is governed by $\|h\|_{\mathcal{H}_k}$ rather than by the norm of G in the raw-input space.*

Proof. See Supplement S8.17.

The statement is for scalar f ; for the vector-valued targets of the campaigns it is applied coordinatewise, with the norm the root sum of squares over output coordinates, which is the norm Theorem 6.9 uses.

The proof (Supplement S8) is the standard pullback identity for reproducing kernels (Saitoh, 1997; Paulsen and Raghupathi, 2016). The identity specifies the feature-kernel RKHS and the minimum extension norm. It provides no comparison with the raw-input RKHS without additional assumptions. In particular, a linear feature map is itself an input metric transformation. The measured fitted-norm reduction is evidence about these fitted kernels at the reported scales and nuggets, rather than a theorem about representation learning.

6.4 An error bound for the residual kernel correction

The final stage of the pipeline is kernel ridge regression of the ensemble residual. The bound below is a vector-valued, residual form of the classical power-function estimate for kernel interpolation (Wendland, 2004) and of the optimal-recovery viewpoint of Owhadi and Scovel (2019); we include the short argument in Supplement S8 to keep the two factors explicit. Fix the mean $m : \mathcal{U} \rightarrow \mathcal{V}$ (in the pipeline, the stacked ensemble; in the statement below m is any realized map; no sample independence is needed for this deterministic statement) and write $r = G - m$ for the residual operator with components r_j , $j = 1, \dots, q$. Let k be a positive definite kernel on $\mathcal{U}$ with RKHS $\mathcal{H}_k$, and assume $r_j \in \mathcal{H}_k$ for all j with

$$\|r\|_K^2 := \sum_{j=1}^q \|r_j\|_{\mathcal{H}_k}^2 < \infty.$$

Given inputs $X = (u_1, \dots, u_n)$, Gram matrix $K = k(X, X)$ and nugget $\lambda \geq 0$, the correction is $\hat{r}_\lambda(u) = k(u, X) (K + n\lambda I)^{-1} R$, $R_{nj} = r_j(u_n)$, and the corrected predictor is $m + \hat{r}_\lambda$. Let

$$P_\lambda(u)^2 = k(u, u) - k(u, X) (K + n\lambda I)^{-1} k(X, u) \quad (4)$$

denote the Gaussian process posterior variance with the same nugget (for $\lambda = 0$ this is the classical power function of the point set X).

Theorem 6.9 (conditional power-function bound). *For every $u \in \mathcal{U}$ and every $\lambda \geq 0$ (for $\lambda = 0$ take K positive definite, or read K^{-1} as the pseudoinverse),*

$$\|G(u) - m(u) - \hat{r}_\lambda(u)\|_2 \leq \|G - m\|_K \tilde{P}_\lambda(u) \leq \|G - m\|_K P_\lambda(u),$$

where $\tilde{P}_\lambda(u)^2 = P_\lambda(u)^2 - n\lambda \|(K + n\lambda I)^{-1} k(X, u)\|_2^2$. The first inequality is sharp: for every u there is a residual operator of norm $\|G - m\|_K$ at which it holds with equality, so $\|G - m\|_K \tilde{P}_\lambda(u)$ is the exact worst-case error of the correction over the ball $\{r : \|r\|_K \leq \|G - m\|_K\}$.

Proof. See Supplement S8.19.

The recorded correction labels. Let $M_{\text{tr}} \in \mathbb{R}^{n \times q}$ denote the training-row mean actually used by the implementation, and let m_{full} be the query-time mean. The kernel channel uses foldwise fits with pooled target centering; neural channels are in-sample, and the refiner's kernel input also changes at deployment. Thus M_{tr} need not be the restriction of a single cross-fitted function. Define

$$c_\lambda(u) = (K + n\lambda I)^{-1} k(X, u), \quad \Delta = m_{\text{full}}(X) - M_{\text{tr}}, \quad R_{\text{tr}} = G(X) - M_{\text{tr}}.$$

The implemented predictor is $m_{\text{full}}(u) + c_\lambda(u)^\top R_{\text{tr}}$. If $G - m_{\text{full}} \in H_k^q$, Theorem 6.9 and the triangle inequality give

$$\begin{aligned} \|G(u) - m_{\text{full}}(u) - c_\lambda(u)^\top R_{\text{tr}}\|_2 &\leq \|G - m_{\text{full}}\|_K \tilde{P}_\lambda(u) + \|c_\lambda(u)^\top \Delta\|_2, \\ \|c_\lambda(u)^\top \Delta\|_2 &\leq \|c_\lambda(u)\|_2 \|\Delta\|_F. \end{aligned} \quad (5)$$

Indeed, $R_{\text{tr}} = (G - m_{\text{full}})(X) + \Delta$; substituting this identity and applying the theorem proves the display. This proof is included here. The mismatch has not been measured, so the single-mean theorem is not an evaluated certificate for the reported correction.

The RKHS assumption is also unverified for these targets and the norm is unknown. For labels sampled from one compatible $r \in H_k^q$, ridge contraction gives $\|\hat{r}_\lambda\|_K \leq \|r\|_K$. With the recorded mixed training labels, the fitted norm need not lower-bound $\|G - m_{\text{full}}\|_K$. Table S6.2 therefore describes fitted functions only: its squared-norm ratio is about 4900 before energy normalization and 8.8 after it. Neither ratio proves a corresponding reduction in an unknown residual norm or smoothness. The design factor is a worst-case error factor and need not rank realized errors well; Section 4.7 measures that ranking directly.

The design factor cannot be improved by a better estimator either. Among all rules that see the same training residuals, the interpolant is optimal in the worst case over the ball, and the price of a positive nugget in that worst case is explicit.

Theorem 6.10 (minimax optimality over the native-space ball). *Fix a design X with K positive definite, a point u and a radius $\rho > 0$. For every map $\Psi : \mathbb{R}^{n \times q} \rightarrow \mathbb{R}^q$ that estimates $r(u)$ from the training residuals $r(X)$,*

$$\sup_{\|r\|_K \leq \rho} \|r(u) - \Psi(r(X))\|_2 \geq \rho P_0(u),$$

and the kernel interpolant $\Psi(R) = k(u, X)K^{-1}R$ attains the bound. For $\lambda > 0$ the worst case of the ridge correction over the same ball is $\rho \tilde{P}_\lambda(u)$, and, with $k_u = k(X, u)$ and $A_\lambda = K + n\lambda I$,

$$P_0(u)^2 \leq \tilde{P}_\lambda(u)^2 \leq P_\lambda(u)^2, \quad \tilde{P}_\lambda(u)^2 - P_0(u)^2 = (n\lambda)^2 k_u^\top A_\lambda^{-2} K^{-1} k_u,$$

so the nugget's cost in the worst-case constant is a computable quantity that vanishes as $\lambda \rightarrow 0$.

Proof. See Supplement S8.20.

The lower bound is the classical optimal-recovery argument (Owhadi and Scovel, 2019): two residuals of norm ρ that vanish on the design and take opposite values at u are indistinguishable from the data, so any rule errs by at least $\rho P_0(u)$ on one of them. What the theorem adds for the pipeline is a reading of the reported quantity: at $\lambda = 0$ the worst-case constant $\|G - m\|_K P_0(u)$ over the native-space ball is the smallest any rule can attain from these data, and at the validated nugget the gap to it is the displayed correction term, evaluable from the Gram matrix alone. The statement is about the worst case over a ball whose radius is not known; it is not an operational error bar for the shipped surrogate, which is the split-conformal set of the next subsection.

6.5 Distribution-free coverage for the reported band

Theorem 6.9 controls the error by $\|G - m\|_K P_\lambda(u)$, but the constant $\|G - m\|_K$ is not known a priori and Section 4.7 shows that P_λ alone ranks the error weakly. What the surrogate reports as uncertainty is therefore a rescaling $q P_\lambda(u)$, whose multiplier is the split-conformal quantile of the scores $\|e(\tilde{u}_i)\|_2 / P_\lambda(\tilde{u}_i)$ on a calibration split of the test block, drawn after every selection in the pipeline has been made on the validation split; Section 4.7 reports this band beside a constant-width band calibrated the same way from the raw scores $\|e(\tilde{u}_i)\|_2$. Coverage does not require the RKHS bound or a useful error ranking. For a predictor and positive scale fixed independently of exchangeable calibration and future inputs, Proposition S3.1 gives marginal coverage at least $1 - \alpha$. Its proof is in Supplement S8. The upper bound $1 - \alpha + 1/(m + 1)$ additionally requires almost surely distinct scores. Define the conformal quantile using the $k = \lceil (1 - \alpha)(m + 1) \rceil$ -th order statistic of the m scores augmented by $+\infty$, which also covers $k = m + 1$.

If, conditional on the fitted predictor and scale, the calibration and evaluation scores are i.i.d. with a continuous distribution and $k \leq m$, the random conditional coverage over calibration draws has law $\text{Beta}(k, m + 1 - k)$. The number covered among N fresh evaluation cases is Beta-binomial. Remark S3.2 states and derives this law in Supplement S3. At $m = 1000$ and $N = 19000$ its nominal-90% central reference interval is $[88.0\%, 91.8\%]$. A random split alone does not establish continuity or undo study-level adaptation to an already

inspected test block. The recorded coverage experiment is compared with this reference law under those conditions; it is not a guarantee conditional on each input or on arbitrary prior test-driven selection.

7 Discussion

The two benchmarks give different roles to a kernel fitted after neural training. On structural mechanics, the neural members and their combination account for most of the accuracy, with a smaller repeatable gain from residual correction. On OCO-2, the representation learned by the network is the main advantage: the direct-target kernel head performs substantially better on frozen features than on raw inputs. Coordinate selection then balances errors across two metrics that weight the reduced outputs very differently.

The structural-mechanics score of $4.546 \pm 0.003\%$ is close to PARA-Net’s published 4.55%, but the available evidence cannot establish a ranking or statistical equivalence between them. The improvement in the lower-data regime is larger, subject to uncertainty about the source study’s exact training subset. These comparisons describe the implementation in Section 3, including its pooled target centering. Replacing that centering with strictly fold-local statistics would change the downstream estimator, and its effect has not been measured.

The ensemble analysis gives a more specific conclusion. For the sixty trained predictors, the empirical lower bound of 4.814% lies close to the hindsight-optimal global convex RMS error of 4.876%. There is little room left for this form of averaging within the pool. Both individual accuracy and residual alignment matter: completing the FNO schedule improves its own error without improving the five-member pipeline, whereas adding the UNet improves the corrected pipeline by 0.032 percentage points. That last result comes from an ablation of the full estimator, whose per-pixel weights and kernel correction extend beyond the class covered by the convex bound. The shared residual is therefore evidence about these predictors, not an irreducible floor for the benchmark.

The OCO-2 comparison benefits from access to the source emulator’s predictions on the same test states. Its scope is the retained forty-component representation: full-spectrum truncation, instrument response, Jacobians, and retrieval accuracy are not evaluated. The coordinate-selected predictor stays close to the best head on the reduced criterion while improving radiance error, rather than dominating every constituent on both. A frozen linear-readout control supports the benefit of the nonlinear kernel head in these experiments. The feature heads select a boundary of the tuning grid, however, and the networks have not converged at the chosen training budget; broader tuning or longer training could change the margins.

The theory helps interpret these results without turning the diagnostics into error bars. The native-space bound separates a residual norm from a design factor and gives the exact worst-case cost of a positive ridge parameter. Applying it to the implemented correction also requires the label-mismatch term in Section 6.4. Neither the residual’s RKHS membership nor an upper bound on its norm has been established for the benchmark target. Fitted norms describe the finite-design regressions, and their ratios do not determine ratios of the unknown target norms. The pullback identity likewise characterizes the space induced by neural features without guaranteeing that its norm is smaller than the raw-input norm.

For uncertainty quantification, the simpler scale works better here. Both kernel power functions and ensemble disagreement rank errors weakly, and a constant-width calibrated set is narrower on average. All three constructions have the same marginal split-conformal guarantee when the predictor and scale are fixed independently of exchangeable calibration and evaluation data. The exact Beta and Beta-binomial reference laws further require i.i.d. continuous scores. Their observed agreement with coverage on the public test block does not protect against selection across studies that repeatedly inspect that block.

Several experiments would sharpen these conclusions. Evaluation on newly generated cases would test whether the measured gains persist beyond the familiar benchmark blocks. Refined-mesh solutions for difficult mechanics cases would help distinguish numerical target effects from shared surrogate bias, and a fold-local-centering rerun would isolate the implementation issue above. On OCO-2, a wider kernel grid, longer training, and a retrieval-level evaluation would establish how far the representation advantage carries. The accompanying supplement provides the proofs and computational details needed to assess the present evidence.

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Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author made substantial use of generative AI tools, and describes their role here. A large part of the software was written with Fable 5 and Fable 5.1 (Anthropic, through the Claude Code agent, May to September 2026): the implementation of the experiments and diagnostics, the downloading, assembly and preparation of the benchmark and OCO-2 datasets, and the adaptation of the publicly released emulator code of Lamminpää et al. — originally split between Julia and a ReFRACtor-driving Python layer — into a single Python pipeline for sampling states and reducing radiances. The study’s contributions are the empirical comparisons, the two-regime framing that organizes the paper, and the supporting theory, all developed with AI assistance in calculation, drafting, and exposition. AI-assisted review was used for mathematical and editorial checks, including

separate reviews of the mathematical arguments and exposition. Selected reported numbers were checked against the released per-run summaries; the training campaigns were not rerun for this revision. The author reviewed all of the above and takes full responsibility for the content of the manuscript.

Data and code availability

The code, the per-run summaries behind every reported number, and the exact configuration of each experiment are available at <https://github.com/yspennstate/neural-means-kernel-corrections>. The base source for this revision was commit 50c05ff1cce3 (full identifier in the source-package README), whose main PDF matches the original supplied manuscript. The repository’s `paper/` directory contains the current article, supplement, and editable sources. Full checkpoints and large prediction/residual arrays are not bundled in the public repository; the source record reports them as retained and available on request. The structural-mechanics, Helmholtz, Navier–Stokes and advection data are distributed through the data record accompanying de Hoop et al. (2022) (data.caltech.edu, record 20091); the OCO-2 emulation data and the kernel-flow emulator’s predictions through the OSF project u2t8a; and the ClimSim data through the LEAP repository `subsampling_low.res`.

Supplementary material

The online supplement, referenced throughout as Supplement S1–S10, is part of this manuscript. The repository hosts the current article and supplement, together with their editable sources in `paper/`. Consult the supplement for the full proofs; each formal result identifies its specific proof location. It contains the finite-sample bounds for every fitted stage and all proofs, the margin and per-coordinate selection results, the effective-dimension identity, the ablation, error-localization, spectra, uncertainty-quantification and cost detail on structural mechanics, the uncontrolled survey of the remaining suite problems and the mirror-symmetry check, the OCO-2 input-metric study with the data-scaling and ClimSim results, the implementation and compute record, and the numerical verification of every stated result. Both documents and their editable sources are also included in the accompanying Overleaf package.

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